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TCP-JSON COMMUNICATION PROTOCOL

Sojet Marking Technology (Xiamen) Co., Ltd

Catalogue

I. Revision History 4

II. Document description 5

1. Introduction 5

2. Format Specification 5

III. Command Structure 5

1. Print control 5

1.1. Start printing 5

1.2. Stop printing 6

1.3. Clear cache 6

1.4. Get printing real-time information 7

1.5. Modify Initial value 8

1.6. Cleaning nozzle 9

2. Printhead parameter 10

2.1. Get printer list 10

2.2. Update printerlist 11

2.3. Get a list of printhead 14

2.4. Update printhead list information 15

3. Printer Setting 16

3.1. Get print settings 16

3.2. Update print setting 17

4. System settings 18

4.1. Get system settings 19

4.2. Updated system settings 19

4.3. Get system time 20

4.4. Set system time 21

4.5. Restore factory settings 21

5. Radix settings 22

5.1. Get Radix list 22

5.2. Add Radix 23

5.3. Find a radix 23

5.4. Delete Radix 24
5.5. Modify radix 24
6. Date formate settings 25
6.1. Get date format list 25
6.2. New date format 26
6.3. Find date format list 27
6.4. Delete Date format 28
6.5. Update date format 29
7. Shift format settings 30
7.1. Get shift list 30
7.2. Creat a shift 31
7.3. Find shift 31
7.4. Delete shift 32
7.5. Edit shift 33
8. Alarm kit 33
8.1. Get alarm kit configuration 33
8.2. Alarm kit Settings 34
8.3. Restore alarm kit settings 35
9. System Status 36
9.1. Get heart beat 36
9.2. Get system Status 37
10. Message operation 39
10.1. Add source 39
10.2. Add object 40
10.3. New message 41
10.4. Find source 42
10.5. Find object 43
10.6. Find message 43
10.7. Get message list 46
10.8. Delete source 47
10.9. Delete object 48
10.10. Delete message 48

10.11. Modify Source 49
10.12. Modift object 49
10.13. Modify message 51
11. Actively report the print status 52
11.1. Actively report the print status 52
12. Send dynamic data 52
12.1. Send dynamic data 52

I. Revision History

Version	Description	Date
1.0		

II. Document description

1. Introduction

1.1. Communication mode

Device communicates with Ethernet port. Establish TCP communication for independent communication.

1.2. Communication Parameters

TCP channel port is 9944

2. Format Specification

2.1. JSON format.

2.2. JSON code cannot have line breaks in the middle, which can no hexadecimal 0D0A.

2.3. The ending must be a newline, which is hexadecimal 0D0A.

III. Command Structure

1. Print control

1.1. Start printing

1.1.1. Data Schema:

Name	Level	Previous level	Data	Type	Description
request_type	1		post	String	Request type, Fixed value
path	1		/engine/printjob	String	Interface path, Fixed value
hash	1		Random number	Interger	Command recognition for consistent sending and receiving
attribute	1			Object	
print_data_name	2	attribute	Message name	string	Message name

1.1.2. Example:

{"request_type":"post","path":"/engine/printjob","hash": 11112,"attribute": {"print_data_name":"Msg" }}

1.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		Ok / Error	status
hash	1		Random number	Command recognition for consistent sending and receiving
attribute	1			Instruction code

create_time	2	attribute	Creation time	Creation time
code	1			
descript	1		Error cause	Describe the cause of the Error

1.1.4. Response Examples

Success return data:

{"status":"ok","hash":11112,"id":949817907,"attribute":{"create_time":949817907}}

Error return data:

{"status":"error","id":949817351,"code":24581,"descript":"print engine is running"}

1.2. Stop printing

1.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/engine/printjob	Interface path, Fixed value
id	1		0	ID, which can be fixed to 0

1.2.2. Example:

{"request_type":"delete","path":"/engine/printjob","id":0}

1.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		Ok or error	Status
id	1			ID
code	1			
descript	1		Error cause	Error cause

1.2.4. Responses Example:

Success return data:

{"status":"ok"}

Error return data:

{"status":"error","id":949817907,"code":24582,"descript":"print engine is stoped"}

1.3. Clear cache

1.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/engine/clear_cache	Interface path, Fixed value

1.3.2. Example

{"request_type": "put","path": "/engine/clear_cache"}

1.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
hash	1		Random number	Command recognition for consistent sending and receiving
id	1			
attribute	1			Instruction code
create_time	2	attribute	Creation time	Creation time
code	1			
descript	1		Error cause	Error cause

1.3.4. Responses Example:

Success return data:

{"status":"ok","hash":0,"id":0,"attribute":{"create_time":949891849}}

Error return data:

{"status":"Error","id":1,"code":24582,"descript":"print engine is stoped"}

1.4. Get printing real-time information

1.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/engine/real	Interface path, Fixed value

1.4.2. Example

{"request_type": "get","path": "/engine/real"}

1.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or error	Status
data_id	1			Message ID
data_name	1			Message name
output	1			Output
ink_used	1		Creation time	Ink Used
start_time	1			Start time
code	1		Error cause	encoding
state	1			
reprint	1		true or false	The state of reprinting

attribute	1			
print_log	2	attribute		
trans_ready	1			
trans_delay	1			
source_info	1		Array	source details, the array contains details of each source.
id	2	source_info		Source ID
type	2	source_info		Source name
content	2	source_info		This source will print the content

1.4.4. Responses Example:

Success return data:

{"status":"ok","data_id":11,"data_name":"Msg2","output":0,"ink_used":0,"start_time":949894275,"code":0,"state":"started","reprint":false,"attribute":{"print_log":false},"trans_ready":true,"trans_delay":[0,0],"source_info":[{"id":4,"type":"counter","name":"count","current":1,"copies_index":1,"content":"00000001","alarm_status":false},{"id":1,"type":"schedule","content":"C"},{"id":1,"type":"date","content":"20000207"}]}

Data returned when printing is not enabled:

{"status":"ok","code":0,"state":"completed","reprint":false}

1.5. Modify Initial value

1.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/engine/parm_modify	Interface path, Fixed value
cmd	1		up_parm:Modify the initial value; reset_parm:Reset parameter; refresh_parm: Refresh parameters. Currently, only updating the current value is enabled	
parm	1		Array	
type	2	parm		Type
id	2	parm		ID
value	2	parm		Data

1.5.2. Example

{"request_type": "post","path": "/engine/parm_modify","cmd": "up_parm","parm": [{"type": "counter","id": 1,"value": 111},{"type": "counter","id": 2,"value": 112}]}

1.5.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
id	1			ID
code	1			
descript	1		Error cause	Error cause

1.5.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":24582,"descript":"print engine is stoped"}

1.6. Cleaning nozzle

1.6.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/engine/printhead_clear	Interface path, Fixed value
id	1		0-3	Printhead ID
columns	1		Number of cleaning rows	Default 50

1.6.2. Example

{"request_type": "put","path": "/engine/printhead_clear", "id": [0,1,2,3],"columns": [200,200,200,200]}

1.6.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status

1.6.4. Responses Example:

Success return data:

{"status":"ok"}

2. Printhead parameter

2.1. Get printer list

2.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/printer	Interface path, Fixed value

2.1.2. Example

{"request_type": "get","path": "/system/printer"}

2.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
printer_list	1		Array	Printer list
id	2	printer_list		ID
name	2	printer_list		Name
printhead_list	2	printer_list	0-3	Printer list
attribute	2	printer_list		
hdpi	3	attribute	Default 300	Horizontal resolution
mirror	3	attribute	True or false	Horizontal mirror or not
reverse	3	attribute	True or false	Vertical mirror or not
nozzle	3	attribute	odd:Left even:Right all:All odd_even:Left/Right super:Speeding	Nozzle
dir	3	attribute	left_to_right:Left to Right right_to_left:Right to left	Print direction
photocell_offset	3	attribute		Distance from Sensor to printhead
auto_clean	3	attribute	never:Never often:Preset custom:Custom	Pre-purge
clean_interval	3	attribute	1-3600	Pre-purge interval
clean_cols	3	attribute	1-1000	Pre-purge Columns
print_mode	3	attribute	single: Single multiple: Multiple	Print mode

			continuous: Continuous	
print_copies	3	attribute	2-255	Copied
print_margin	3	attribute	0-9999.99	Interval
iclean	3	attribute	True or false	
iclean_interval	3	attribute		
auto_print_copies	3	attribute		
ink_control_count	3	attribute		
ink_control	3	attribute		
photocell	3	attribute		

2.1.4. Responses Example:

Success return data:

{"status":"ok","printer_list":[{"id":0,"name":"Printer1","printhead_list":[0,1],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":8,"ink_control_count":7,"ink_control":"none","photocell":0}},{"id":1,"name":"Printer2","printhead_list":[],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":6,"ink_control_count":7,"ink_control":"none","photocell":0}}]}

2.2. Update printerlist

2.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/printer	Interface path, Fixed value
printer_list	1		Array	Printer list
id	2	printer_list		Id
name	2	printer_list		name
printhead_list	2	printer_list	0-3	Printhead list
attribute	2	printer_list		
hdpi	3	attribute		Horizontal resolution
mirror	3	attribute	True or false	Horizontal mirror or not
reverse	3	attribute	True or false	Vertical mirror or not
nozzle	3	attribute	odd:Left even:Right all:All odd_even:Left/Right super:Speeding	Nozzle

dir	3	attribute	left_to_right:Left to Right right_to_left:Right to left	Print direction
photocell_offset	3	attribute		Distance from Sensor to printhead
auto_clean	3	attribute	never:Never often:Preset custom:Custom	Pre-purge
clean_interval	3	attribute	1-3600	Pre-purge interval
clean_cols	3	attribute	1-1000	Pre-purge Columns
print_mode	3	attribute	single: Single multiple: Multiple continuous: Continuous	Print mode
print_copies	3	attribute	2-255	Copied
print_margin	3	attribute	0-9999.99	Interval
iclean	3	attribute	True or false	
iclean_interval	3	attribute		
auto_print_copies	3	attribute		
ink_control_count	3	attribute		
ink_control	3	attribute		
photocell	3	attribute		

2.2.2. Example

{"request_type": "put","path": "/system/printer","printer_list":[{"id":0,"name":"Printer1","printhead_list":[0,1],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":8,"ink_control_count":7,"ink_control":"none","photocell":0}},{"id":1,"name":"Printer2","printhead_list":[],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":6,"ink_control_count":7,"ink_control":"none","photocell":0}}]}

2.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
printer_list	1		Array	Printer list
id	2	printer_list		ID
name	2	printer_list		name
printhead_list	2	printer_list	0-3	Printhead list

attribute	2	printer_list		
hdpi	3	attribute	Default 300	Horizontal resolution
mirror	3	attribute	True or false	Horizontal mirror or not
reverse	3	attribute	True or false	Vertical mirror or not
nozzle	3	attribute	odd:Left even:Right all:All odd_even:Left/Right super:Speeding	Nozzle
dir	3	attribute	left_to_right:Left to Right right_to_left:Right to left	Print direction
photocell_offset	3	attribute		Distance from Sensor to printhead
auto_clean	3	attribute	never:Never often:Preset custom:Custom	Pre-purge
clean_interval	3	attribute	1-3600	Pre-purge interval
clean_cols	3	attribute	1-1000	Pre-purge Columns
print_mode	3	attribute	single: Single multiple: Multiple continuous: Continuous	Print mode
print_copies	3	attribute	2-255	Copied
print_margin	3	attribute	0-9999.99	Interval
iclean	3	attribute	True or false	
iclean_interval	3	attribute		
auto_print_copies	3	attribute		
ink_control_count	3	attribute		
ink_control	3	attribute		
photocell	3	attribute		

2.2.4. Responses Example:

Success return data:

{"status":"ok","printer_list":[{"id":0,"name":"Printer1","printhead_list":[0,1],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":8,"ink_control_count":7,"ink_control":"none","photocell":0}},{"id":1,"name":"Printer2","printhead_list":[],"attribute":{"hdpi":300,"mirror":false,"reverse":false,"nozzle":"all","dir":"left_to_right","photocell_offset":0,"auto_clean":"never","clean_interval":10,"clean_cols":50,"print_mode":"single","print_copies":2,"print_margin":4,"iclean":false,"iclean_interval":120,"auto_print_copies":6,"ink_control

_count":7,"ink_control":"none","photocell":0}}]}

2.3. Get a list of printhead

2.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/printhead_list	Interface path, Fixed value

2.3.2. Example

{"request_type": "get","path": "/system/ printhead_list "}

2.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
printhead_list	1		Array	Printhead list
id	2	printhead_list		Printhead id
name	2	printhead_list		Printhead name
attribute	2	printhead_list		
offset_x	3	attribute		X offset Unit: point
offset_y	3	attribute		Y offset Unit: point
intrinsic_distance_x	3	attribute		printhead Unit: mm
intrinsic_offset_y	3	attribute		Print head vertical offset unit: point
firing_voltage	3	attribute		Voltage unit: V
pulse_width	3	attribute		Pulse width unit:us
temp_control	3	attribute		Temperature control switch
total_prints	3	attribute		Total prints
preheat_temp	3	attribute		Residual temp
preheat_loop_temp	3	attribute		Preheat Loop Temp
residual_temp	3	attribute		Remaining time
description	3	attribute		
resolution	3	attribute		
nozzle_count	3	attribute		
residual_temp_multiplier	3	attribute		
residual_temp_control	3	attribute		
ink_price	3	attribute		

2.3.4. Responses Example:

Success return data:

{"printhead_list":[{"id":0,"name":"P1","attribute":{"offset_x":0,"offset_y":0,"intrinsic_distance_x":33.200001,"intrinsic_offset_y":2,"description":"printhead0","resolution":600,"nozzle_count":300,"firing_voltage":8.800000,"pulse_width":1.800000,"preheat_temp":39,"preheat_loop_temp":0,"residual_temp":42.187500,"residual_temp_multiplier":1,"ink_price":0,"temp_control":false,"residual_temp_control":false,"total_prints":"67"}},{"id":1,"name":"P2","attribute":{"offset_x":0,"offset_y":0,"intrinsic_distance_x":33.200001,"intrinsic_offset_y":2,"description":"printhead1","resolution":600,"nozzle_count":300,"firing_voltage":8.400000,"pulse_width":1.800000,"preheat_temp":39,"preheat_loop_temp":0,"residual_temp":42.187500,"residual_temp_multiplier":1,"ink_price":0,"temp_control":false,"residual_temp_control":false,"total_prints":"67"}}]}

2.4. Update printhead list information

2.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/printhead_list	Interface path, Fixed value
printhead_list	1		Array	Printhead list
id	2	printhead_list		Printhead id
name	2	printhead_list		Printhead name
attribute	2	printhead_list		
offset_x	3	attribute		X offset Unit: point
offset_y	3	attribute		Y offset Unit: point
intrinsic_distance_x	3	attribute		printhead Unit: mm
intrinsic_offset_y	3	attribute		Print head vertical offset unit: point
firing_voltage	3	attribute		Voltage unit: V
pulse_width	3	attribute		Pulse width unit:us
temp_control	3	attribute		Temperature control switch
total_prints	3	attribute		Total prints
preheat_temp	3	attribute		Residual temp
preheat_loop_temp	3	attribute		Preheat Loop Temp
residual_temp	3	attribute		Remaining time

2.4.2. Example

{"request_type": "put","path": "/system/printhead_list","printhead_list":[{"id":0,"name":"P1","attribute":{"offset_x":0,"offset_y":0,"intrinsic_distance_x":33.200001,"intrinsic_offset_y":2,"firing_voltage":8.800000,"pulse_width":1.800000,"preheat_temp":39,"preheat_loop_temp":0,"residual_temp":42.187500}},{"id":1,"name":"P2","attribute":{"offset_x":0,"offset_y":0,"intrinsic_distance_x":33.200001,"intrinsic_offset_y":2,"firing_voltage":8.400000,"pulse_w

idth":1.800000,"preheat_temp":39,"preheat_loop_temp":0,"residual_temp":42.187500}}]}

2.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

2.4.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

3. Printer Setting

3.1. Get print settings

3.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/print_settings	Interface path, Fixed value

3.1.2. Example

{"request_type": "get","path": "/system/print_settings"}

3.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
Id	1			ID
name	1			Name
description	1			Description
photocell_type	1			Photocell type
photocell	1			Built-in sensor
encoder_type	1			Encoder type
hdpi	1			Horizontal resolution
vdpi	1			Vertical resolution
encoder_speed	1			Production line speed
encoder_resolution	1			Encoder resolution
photocell_sense	1			Photocell sense
photocell_offset	1			Sensor distance

history_limit	1			History Limit
cache_limit	1			Cache Limit
print_dialog	1			Reset counter
ambient_temp	1			
print_media_width	1			Print media width
trigger_distance	1			
printhead_print_swath	1			
photocell_filter_time	1			Photocell filiter time
low_ink_level	1			
photocell_filter_count	1			
realtime_printing	1			Real-time print
roundtrip_printing	1			Roundtrip print
nt	1			Whether the mode is NT
q_photocell	1			Q photocell
speed_meas	1			Automatic speed switch
auto_printer_mode	1			
db_potocell_register	1			
potocell_filter_register	1			
info_scan_settings	1			Special scan

3.1.4. Responses Example:

Success return data:

{"id":0,"name":"print","description":"","photocell_type":"peripheral","photocell":"P1","encoder_type":"2phase","hdpi":600,"vdpi":600,"encoder_speed":60,"encoder_resolution":300,"photocell_sense":"edge","photocell_offset":600,"history_limit":10,"cache_limit":10,"print_dialog":0,"ambient_temp":25,"print_media_width":0,"trigger_distance":120,"printhead_print_swath":12.700000,"photocell_filter_time":0,"low_ink_level":95,"photocell_filter_count":0,"realtime_printing":false,"roundtrip_printing":false,"nt":false,"q_photocell":true,"speed_meas":false,"auto_printer_mode":"none","db_potocell_register":0,"potocell_filter_register":0,"info_scan_settings":[]}

3.2. Update print setting

3.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value

path	1		/system/printhead_list	Interface path, Fixed value
id	1			ID
name	1			Name
description	1			Description
photocell_type	1			Photocell type
photocell	1			Photocell
encoder_type	1			Encoder type
encoder_speed	1			Production line speed
encoder_resolution	1			Encoder resolution
photocell_sense	1			Photocell sense
history_limit	1			History Limit
cache_limit	1			Cache Limit
print_dialog	1			Print log
realtime_printing	1			Realtimne print
roundtrip_printing	1			Roundtrip print

3.2.2. Example

{"request_type": "put","path": "/system/print_settings","id": 0,"name": "print","description": " ","photocell_type": "peripheral","photocell": "P1", "encoder_type": "2phase","encoder_speed": 60,"encoder_resolution": 300,"photocell_sense": "edge","history_limit": 10,"cache_limit": 10,"print_dialog": 0,"realtime_printing": false,"roundtrip_printing": false}

3.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

3.2.4. Responses Example:

Success return:

{"status":"ok","code":0,"descript":"ok"}

4. System settings

4.1. Get system settings

4.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/system_settings	Interface path, Fixed value

4.1.2. Example

{"request_type": "get","path": "/system/system_settings"}

4.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
screen_off_timeout	1			Screen saver
brightness	1			Brightness
device_name	1			Device name
font	1			Font
measure_system	1			Unit
ip_assignment	1			Networkmode
address	1			IP
netmask	1			Gateway
gateway	1			Netmask
hardware_version	1			Hardware version
mcu_version	1			Mcu version
fpga_version	1			Fpga version
serial_number	1			SN
client_code	1			Printer code
server_version	1			Software version
version	1			Version
date	1			

4.1.4. Responses Example:

Success return data:

{"screen_off_timeout":28800000,"brightness":255,"device_name":"MoEco 2000","font":"default.ttf","measure_system":"metric","ip_assignment":"dhcp","address":"192.168.0.100","netmask":"255.255.255.0","gateway":"192.168.0.1","hardware_version":"2e","mcu_version":"1.25","fpga_version":"1.25","serial_number":"0","client_code":"1459257713","server_version":"1.4.0-20231107105855","version":"1.4.1","date":"20231107110302"}

4.2. Updated system settings

4.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/system_settings	Interface path, Fixed value
screen_off_timeout	1			Screen saver
brightness	1			Brightness
device_name	1			Device name
font	1			Font
measure_system	1			Unit
ip_assignment	1			Networkmode
address	1			IP
netmask	1			Netmask
gateway	1			Gateway

4.2.2. Example

{"request_type": "put","path": "/system/system_settings","screen_off_timeout": 28800000,"brightness": 155,"device_name": "MoTix 4000","font": "default.ttf","locale": "zh-CN","measure_system": "metric","ip_assignment": "static","address": "192.168.0.100","netmask": "255.255.255.0", "gateway": "192.168.0.1"}

4.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

4.2.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

4.3. Get system time

4.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/date_get	Interface path, Fixed value

4.3.2. Example

{"request_type": "get","path": "/system/date_get"}

4.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
year	1			year
month	1			month
day	1			day
hour	1			hour
minute	1			minute
second	1			second

4.3.4. Responses Example:

Success return:

{"year": 2023,"month": 10,"day": 1,"hour": 8,"minute": 36,"second": 33}

4.4. Set system time

4.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/date_get	Interface path, Fixed value
year	1			year
month	1			month
day	1			day
hour	1			hour
minute	1			minute
second	1			second

4.4.2. Example

{"request_type": "put","path": "/system/date_set", "year": 2023,"month": 10,"day": 1,"hour": 10,"minute": 30, "second": 41}

4.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

4.4.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

4.5. Restore factory settings

4.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/reset	Interface path, Fixed value
printdata	1		true or false	Message
image	1		true or false	Logo
print_settings	1		true or false	Print settings
system_settings	1		true or false	System settings
language	1		true or false	Language
font	1		true or false	Font
inputmethod	1		true or false	Keyboard

4.5.2. Example

{"request_type": "get","path": "/system/reset"}

4.5.3. Responses Data Schema

Direct recovery, no return.

5. Radix settings

5.1. Get Radix list

5.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
offset	1			Start bit
num	1			Total number

5.1.2. Example

{"request_type": "get","path": "/system/radix_list","offset": 0,"num": 10}

5.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
radix_list	1			Radix list
id	1			ID
name	1			Radix name
radix_digits	1			Radix content

5.1.4. Responses Example:

Success return data:

{"radix_list":[{"id":1,"name":"dec","radix_digits":"0123456789"}]}

5.2. Add Radix

5.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
hash	1			Random number
name	1			Radix name
radix_digits	1			Radix content

5.2.2. Example

{"request_type": "post","path": "/system/radix","hash": 1111222,"name": "radix1","radix_digits": "01"}

5.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
hash	1			Random number
id	1			Add radix id

5.2.4. Responses Example:

Success return data:

{"status":"ok","hash":1111222,"id":2}

5.3. Find a radix

5.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
id	1			Find radix id

5.3.2. Example

{"request_type": "get","path": "/system/radix","id": 2}

5.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
id	1			Find radix id
name	1			Radix name
radix_digits	1			Radix content
status	1		Error	Status
code	1			
descript	1		ok	conclusion

5.3.4. Responses Example:

Success return data:

{"id":2,"name":"radix1","radix_digits":"01"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

5.4. Delete Radix

5.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
id	1			Radix id

5.4.2. Example

{"request_type": "delete","path": "/system/radix_list","id": 2}

5.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

5.4.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

5.5. Modify radix

5.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
id	1			Radix id
name	1			Radix name
radix_digits	1			Radix content

5.5.2. Example

{"request_type": "put","path": "/system/radix","id": 1,"name": "scal1","radix_digits": "0123456789"}

5.5.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

5.5.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

6. Date formate settings

6.1. Get date format list

6.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/dateformat_list	Interface path, Fixed value
offset	1			Start bit
num	1			Total number

6.1.2. Example

{"request_type": "get","path": "/system/dateformat_list","offset": 0,"num": 10}

6.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
dateformat_list	1		Array	Date format list
id	2	Dateformat_list		ID
name	2	dateformat_list		Name
items	2	dateformat_list	Array	
type	3	items		Type Date:date format text: Character string
content	3	items		Content
radix	2	dateformat_list	Array	Radix
name	3	radix		Name
radix_digits	3	radix		Radix Content

6.1.4. Responses Example:

Success return data:

{"dateformat_list":[{"id":1,"name":"YYYYMMDD","items":[{"type":"date","content":"YYYY"},{"type":"date","content":"MM"},{"type":"date","content":"DD"}],"radix":{"name":"dec","radix_digits":"0123456789"}},{"id":2,"name":"YYYY/MM/DD","items":[{"type":"date","content":"YYYY"},{"type":"text","content":"/"},{"type":"date","content":"MM"},{"type":"text","content":"/"},{"type":"date","content":"DD"}],"radix":{"name":"dec","radix_digits":"0123456789"}},{"id":3,"name":"MM/DD/YYYY","items":[{"type":"date","content":"MM"},{"type":"text","content":"/"},{"type":"date","content":"DD"},{"type":"text","content":"/"},{"type":"date","content":"YYYY"}],"radix":{"name":"dec","radix_digits":"0123456789"}},{"id":4,"name":"HH:mm","items":[{"type":"date","content":"HH"},{"type":"text","content":":"},{"type":"date","content":"mm"}],"radix":{"name":"dec","radix_digits":"0123456789"}}]}

6.2. New date format

6.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/system/dateformat	Interface path, Fixed value
hash	1			Random number
name	1			Date format name
items	1		Array	
type	2	items		Type Date:Date formate text:Character string
content	2	items	YYYY:Year; *2021* YYY: Year Last three digits; *021*	Content

			YY:Year last two y digits; *21* Y:Year last digit; *1* MM:Mouth，Numerical representation 1-12 *5* MEN:Mouth，abbreviation *MAY* MALL:Mouth,Full name	
radix	1		Array	Radix
name	2	radix		Name
radix_digits	2	radix		Radix content

6.2.2. Example

{"request_type": "post","path": "/system/dateformat","hash": 123456, "name": "dateformat1","radix": {"name": "radix","radix_digits": "01" },"items": [{"type": "date","content": "YYYY"}]}

6.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
hash				Random number
id	1			Date format id

6.2.4. Responses Example:

Success return data:

{"status":"ok","hash":123456,"id":5}

6.3. Find date format list

6.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/radix_list	Interface path, Fixed value
id	1			Date format id

6.3.2. Example

{"request_type": "get","path": "/system/dateformat","id": 5}

6.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
id	1			Date format id
name	1			Date dormat name
items	1		Array	
type	2	items		Type date:Dateformat text: Character string
content	2	items		Content
radix	1		Array	Radix
name	2	radix		Name
radix_digits	2	radix		Radix content
status	1		Error	Status
code	1			
descript	1		ok	conclusion

6.3.4. Responses Example:

Success return data:

{"id":5,"name":"dateformat1","items":[{"type":"date","content":"YYYY"}],"radix":{"name":"radix","radix_digits":"01"}}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

6.4. Delete Date format

6.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/system/dateformat	Interface path, Fixed value
id	1			Date format id

6.4.2. Example

{"request_type": "delete","path": "/system/dateformat","id": 5}

6.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

6.4.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

6.5. Update date format

6.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/dateformat	Interface path, Fixed value
id	1			Date format id
name				Date format name
items	1		Array	
type	2	items		Type date:Dateformat text: Character string
content	2	items	YYYY:Year; *2021* YYY: Year Last three digits; *021* YY:Year last two ydigits; *21* Y:Year last digit; *1* MM:Mouth，Numerical representation 1-12 *5* MEN:Mouth，abbreviation *MAY* MALL:Mouth,Full name	Content
radix	1		Array	Radix
name	2	radix		Name
radix_digits	2	radix		Radix content

6.5.2. Example

{"request_type": "put","path": "/system/dateformat","id": 6,"name": "dateformat1","radix": {"name": "dec","radix_digits": "0123456789" },"items": [{"type": "date","content": "YYYY"}]}

6.5.3. Responses Data Schema

Name	Level	Previous	Data	Description

		level		
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

6.5.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

7. Shift format settings

7.1. Get shift list

7.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/schedule_list	Interface path, Fixed value
offset	1			Start bit
num	1			Total number

7.1.2. Example

{"request_type": "get","path": "/system/schedule_list","offset": 0,"num": 10}

7.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
schedule_list	1		Array	Shift list
id	2	schedule_list		ID
name	2	schedule_list		Name
shifts	2	schedule_list	Array	Shifts
name	3	shifts		Shift content
time	3	shifts		Time

7.1.4. Responses Example:

Success return data:

{"schedule_list":[{"id":1,"name":"default","shifts":[{"name":"A","time":480},{"name":"B","time":960},{"name":"C","time":0}]}]}

7.2. Creat a shift

7.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/system/schedule	Interface path, Fixed value
hash	1			Random number
name	1			Shift name
shifts	1		Array	shift
name	2	shifts		Name
time	2	shifts		Time

7.2.2. Example

{"request_type": "post","path": "/system/schedule","hash": 1111222,"name": "schedule1","shifts": [{"name": "A","time": 0},{"name": "B","time": 480 },{"name": "C","time": 960}]}

7.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
hash	1			Random number
id	1			Shift id

7.2.4. Responses Example:

Success return data:

{"status":"ok","hash":1111222,"id":2}

Error return data:

{"status":"Error","code":20518,"descript":"Abort due to constraint violation "}

7.3. Find shift

7.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/schedule	Interface path, Fixed value
id	1			Shift id

7.3.2. Example

{"request_type": "get","path": "/system/schedule","id": 2}

7.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
id	1			Shift id
name	1			Shift name
shifts	1		Array	Shift
name	2	shifts		Shift content
time	2	shifts		Tome
status	1		Error	Status
code	1			
descript	1		ok	conclusion

7.3.4. Responses Example:

Success return data:

{"id":2,"name":"schedule1","shifts":[{"name":"A","time":0},{"name":"B","time":480},{"name":"C","time":960}]}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

7.4. Delete shift

7.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/system/schedule	Interface path, Fixed value
id	1			Shift id

7.4.2. Example

{"request_type": "delete","path": "/system/schedule","id": 2}

7.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

7.4.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

7.5. Edit shift

7.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/schedule	Interface path, Fixed value
id	1			Shift id
name	1			Shift name
shifts	1		Array	shift
name	2	shifts		Shift content
time	2	shifts		Time

7.5.2. Example

{"request_type": "put","path": "/system/schedule","id": 3,"name": "schedule1","shifts": [{"name": "A","time": 4800}]}

7.5.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

7.5.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

8. Alarm kit

8.1. Get alarm kit configuration

8.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/system/signal_config	Interface path, Fixed

				value

8.1.2. Example

{"request_type": "get","path": "/system/signal_config"}

8.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
signals	1		Array	Signal
name	2	signals	stop_print ink_empty ink_warning system_Error	name Stop Print, Ink Low, Ink Empty, System Error
icon	2	signals		Icon
handlers	1		Array	Operate
name	2	handlers	turn_on_red_led turn_on_yellow_led beep	Name corresponding to: red light, yellow light, buzz
icon	2	handlers		Icon
signal_config	1		Array	Signal configuration
signal	2	signal_config		Signal
handlers	2	signal_config		operator

8.1.4. Responses Example:

Success return data:

{"signals":[{"name":"stop_print","icon":"stop_print.icon"},{"name":"ink_empty","icon":"ink_empty.icon"},{"name":"ink_warning","icon":"ink_warning.icon"},{"name":"system_Error","icon":"system_Error.icon"},{"name":"network_Error","icon":"network_Error.icon"}],"handlers":[{"name":"turn_on_red_led","icon":"red_on.icon"},{"name":"turn_on_yellow_led","icon":"yellow_on.icon"},{"name":"beep","icon":"beep.icon"}],"signal_config":[{"signal":"stop_print","handlers":[]},{"signal":"ink_empty","handlers":["turn_on_red_led","beep"]},{"signal":"ink_warning","handlers":["turn_on_yellow_led"]},{"signal":"system_Error","handlers":["turn_on_red_led","beep"]},{"signal":"network_Error","handlers":[]}]}

8.2. Alarm kit Settings

8.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/system/signal_config	Interface path, Fixed value
signals	1		Array	Signal
name	2	signals	stop_print ink_empty ink_warning	name Stop Print, Ink Low, Ink Empty, System Err

			system_Error	or
icon	2	signals		Icon
handlers	1		Array	Operate
name	2	handlers	turn_on_red_led turn_on_yellow_led beep	Name corresponding to: red light, yellow light, buzz
icon	2	handlers		Icon
signal_config	1		Array	Signal configuration
signal	2	signal_config		Signal
handlers	2	signal_config		operator

8.2.2. Example

{"request_type": "put","path": "/system/signal_config","signals": [{"name": "stop_print","icon": "stop_print.icon"},{"name": "ink_empty","icon": "ink_empty.icon"},{"name": "ink_warning","icon": "ink_warning.icon"},{"name": "system_Error","icon": "system_Error.icon"}],"handlers": [{"name": "turn_on_red_led","icon": "red_on.icon"},{"name": "turn_on_yellow_led","icon": "yellow_on.icon"},{"name": "beep","icon": "beep.icon"}],"signal_config": [{"signal": "stop_print","handlers": []},{"signal": "ink_empty","handlers": ["turn_on_red_led","beep"]},{"signal": "ink_warning","handlers": ["turn_on_yellow_led"]},{"signal": "system_Error","handlers": ["turn_on_red_led","beep"]}]}

8.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

8.2.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12289,"descript":"arguement Error"}

8.3. Restore alarm kit settings

8.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/system/signal_config	Interface path, Fixed

				value

8.3.2. Example

{"request_type": "delete","path": "/system/signal_config"}

8.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

8.3.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

9. System Status

9.1. Get heart beat

9.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/info/heart_beat	Interface path, Fixed value

9.1.2. Example

{"request_type": "get","path": "/info/heart_beat"}

9.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

9.1.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

9.2. Get system Status

9.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/info/status	Interface path, Fixed value

9.2.2. Example

{"request_type": "get","path": "/info/status"}

9.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1			Status
network	2	status		Network
encoder	2	status		Encoder
usb	2	status		USB
printhead	2	status		Printhead
id	3	printhead		ID
attribute	3	printhead		
software_version	4	attribute		Software version
fpga_version	4	attribute		Fpga version
hardware_version	4	attribute		Hardware version
total_prints	4	attribute		Total printable
status	4	attribute	ok:normal missing:No ink incorrect:Incorrect ink very_low:Ink warning empty:Ink empty	Status
serial_number	4	attribute		Serial number
expiration	4	attribute		expiration
client_code	4	attribute		Printer code
ink_type	4	attribute		Ink type
drop_volume	4	attribute		Drop volume
ink_capacity	4	attribute		Ink capacity
ink_volume	4	attribute		Amount of ink used
firing_voltage	4	attribute		Voltage
pulse_width	4	attribute		Width
temp_control	4	attribute		Temperature control

temp_status	4	attribute		Temperature Status
preheat_temp	4	attribute		Preheating Temperature
preheat_loop_temp	4	attribute		Residual Temperature Multipier
temp	4	attribute		Temperature
residual_temp	4	attribute		Residual Temperature
remaining_output	4	attribute		Remaining printable
total_printable	4	attribute		Total printable
visor_info	2	status		Dust cover Status
id	3	visor_info		Channel ID
enable	3	visor_info		
visor_status	3	visor_info	Yes: true No: false	Enable or not
no_ink_open_enable	3	visor_info		No cartridge open lid
auto_cover_enable	3	visor_info		Automatic closing
auto_cover_time	3	visor_info	5-65535	Automatic closing time
hard_version	2	status		Hardware version
mcu_version	2	status		Mcu version
fpga_version	2	status		Fpga version
serial_number	2	status		Serial number
client_code	2	status		Printer code
photocell	2	status		Sensor status
photocell_onoff	2	status		Sensor status test
encoder_speed	2	status		Encoder speed
test_speed	2	status		Test encoder speed
mobile_print_status	2	status	connected:connected disconnected:Disconnected	Moving printing Status
test_encoder	2	status		Encoder test

9.2.4. Responses Example:

Success return data:

{"status":{"network":"connected","encoder":"connected","usb":"disconnected","printhead":[{"id":0,"attribute":{"software_version":281,"fpga_version":281,"hardware_version":256,"total_prints":"0","status":"ok","serial_number":"2","expiration":1847404801,"client_code":"1459257713","ink_type":"531S","drop_volume":35,"ink_capacity":"1200000000","ink_volume":"965759882","firing_voltage":8.800000,"pulse_width":1.800000,"temp_control":false,"temp_status":"normal","temp":0,"residual_temp":42.187500,"residual_temp_multiplier":1,"remaining_output":0,"total_printable":0}},{"id":1,"attribute":{"software_version":281,"fpga_version":281,"hardware_version":256,"total_prints":"0","status":"ok","serial_number":"2221024008","expiration":1824336001,"client_code":"1459257713","ink_type":"53S+","drop_volume":43,"ink_capacity":"976744186","ink_volume":"15960270","firing_voltage":8.400000,"pulse_width":1.8

00000,"temp_control":false,"temp_status":"normal","temp":0,"residual_temp":42.187500,"residual_temp_multiplier":1,"remaining_output":0,"total_printable":0}}],"visor_info":[{"id":0,"enable":false,"visor_status":false,"no_ink_open_enable":true,"auto_cover_enable":false,"auto_cover_time":0},{"id":1,"enable":false,"visor_status":false,"no_ink_open_enable":true,"auto_cover_enable":false,"auto_cover_time":0}],"hard_version":256,"mcu_version":281,"fpga_version":281,"serial_number":0,"client_code":0,"photocell":"connected","photocell_1":"connected","photocell_onoff":"on","photocell_onoff_1":"on","total_photocell_triggers":0,"total_photocell_triggers_1":0,"encoder_speed":0,"mobile_print_status":"disconnected","test_speed":0,"test_encoder":"pulse1: 0 pulse2: 0 \n curr: 0 accumulate: 0 photocell_num: 0\n finish: no\n"}}

10. Message operation

10.1. Add source

10.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/data/source	Interface path, Fixed value
hash	1			Random number
type	1			Type
name	1			Name
attribute	1			
content	2	attribute		Content

10.1.2. Example

{"request_type": "post","path": "/data/source","hash": 86,"type": "text","name": "text","attribute": {"content": "TXT"}}

10.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
hash	1			Random number
id	1		ok	New source id

10.1.4. Responses Example:

Success return data:

{"status":"ok","hash":86,"id":1}

10.2. Add object

10.2.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/data/source	Interface path, Fixed value
hash	1			Random number
type	1			Type
name	1			Name
style	1			Style
x	2	style		X position
y	2	style		Y position
w	2	style		Width
h	2	style		Height
pivot_x	2	style		X pivot
pivot_y	2	style		Y pivot
rotate	2	style		Rotation
scale_x	2	style		X scale
scale_y	2	style		Y scale
paint_style	2	style		Paint style
line_cap	2	style		
line_join	2	style		
line_width	2	style		
miter_limit	2	style		
font_style	2	style		Font style
visiblity	2	style		Visiblity
attribute	1			
source_list	1		Array	Source list
type	2	source_list		Source type
id	2	source_list		Source id

10.2.2. Example

{"request_type": "post","path": "/data/object","hash": 16381920,"type": "text","name": "text","style": {"x": 0,"y": 23,"w": 777,"h": 110,"pivot_x": 0,"pivot_y": 0,"rotate": 0.0,"scale_x": 1.0,"scale_y": 1.0,"paint_style": "fill","line_cap": "butt","line_join": "miter","line_width": 0, "miter_limit": 4,"font_style": "ttf-default-r*nnn*-378-378-UTF-8","visiblity": "visible"},"attribute": {},"source_list": [{"type": "text","id": 1 }]}

10.2.3. Responses Data Schema

Name	Level	Previous	Data	Description

		level		
status	1		ok or Error	Status
hash	1			Random number
id	1		ok	Source id

10.2.4. Responses Example:

Success return data:

{"status":"ok","hash":16381920,"id":1}

10.3. New message

10.3.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/data/data	Interface path, Fixed value
hash	1			Random number
type	1			type
name	1			name
attribute	1			
printdata_pref	2	attribute		delay
print_prefs_pref	3	printdata_pref		
ff_margin	4	print_prefs_pref		Front delay
fr_margin	4	print_prefs_pref		Rear delay
bf_margin	4	print_prefs_pref		Roundtrip reverse-Front delay
br_margin	4	print_prefs_pref		Roundtrip reverse –Rear delay
object_list	1			Object list
id	2	object_list		Object id
type	2	object_list		Object type

10.3.2. Example

{"request_type": "post","path": "/data/data","hash": 17957816,"name": "Msg","attribute": {"printdata_pref": {"print_prefs": [{"ff_margin": 60.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0},{ "ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0},{"ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0},{"ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin":

0.0}]}},"object_list": [{"id": 1,"type": "text"}]}

10.3.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
id	1			Message id
hash	1			Random number
attribute	1			
created_time	2	attribute		Creation time
modified_time	2	attribute		Modified time

10.3.4. Responses Example:

Success return data:

{"status":"ok","id":1,"hash":17957816,"attribute":{"created_time":1696228686,"modified_time":1696228686}}

10.4. Find source

10.4.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/data/source	Interface path, Fixed value
id	1			Source id
type	1			Source type

10.4.2. Example

{ "request_type": "get","path": "/data/source","id":1,"type":"text"}

10.4.3. Responses Data Schema

Name	Level	Previous level	Data	Description
type	1			Type
id	1			Source id
name	1			Name
attribute	1			Details, different sources return different

10.4.4. Responses Example:

Success return data:

{"type":"text","id":1,"name":"text","attribute":{"content":"TXT","exported":false}}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.5. Find object

10.5.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/data/object	Interface path, Fixed value
id	1			Object id

10.5.2. Example

{"request_type": "get","path": "/data/object","id":1}

10.5.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1			Status
name	1			Name
id	1			ID
locked	1			
lock_aspect_ratio	1			
relative_resize	1			
enabled	1			
style_id	1			Style ID
type	1			Type
source_list	1	source_list		Source list
type	2			Type
id	2			ID

10.5.4. Responses Example:

Success return data:

{"status":"ok","name":"text","id":1,"locked":false,"lock_aspect_ratio":true,"relative_resize":true,"enabled":true,"style_id":1,"type":"text","source_list":[{"type":"text","id":1}]}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.6. Find message

10.6.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/data/data	Interface path, Fixed value
id	1			Message id
detail				

10.6.2. Example

{"request_type": "get","path": "/data/data","id":1,"detail":1}

10.6.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1			Status
id	1			ID
name	1			Name
attribute	1			
created_time	2	attribute		Creation time
modified_time	2	attribute		Modified time
permission	2	attribute		Permission
print_log	2	attribute		
printdata_pref	2	attribute		Delay
print_prefs_pref	3	printdata_pref		
ff_margin	4	print_prefs_pref		Front delay
fr_margin	4	print_prefs_pref		Rear delay
bf_margin	4	print_prefs_pref		Roundtrip reverse-Front delay
br_margin	4	print_prefs_pref		Roundtrip reverse –Rear delay
continuous_print	4	print_prefs_pref		Continuous print
style	1			Style
rotate	2	style		
mirror	2	style		
stretch	2	style		
reverse	2	style		
object_list	1			
type	2	object_list		
name	2	type		

id	2	type		
attribute	2	type		
locked	3	attribute		
lock_aspect_ratio	3	attribute		
relative_resize	3	attribute		
enabled	3	attribute		
style	2	object_list		
font_style	3	style		
rotate	3	style		
mirror	3	style		
stretch	3	style		
reverse	3	style		
visiblity	3	style		
x	3	style		
y	3	style		
w	3	style		
h	3	style		
halign	3	style		
valign	3	style		
row_space	3	style		
letter_space	3	style		
fh_ratio	3	style		
fw_ratio	3	style		
pivot_x	3	style		
pivot_y	3	style		
scale_x	3	style		
scale_y	3	style		
paint_style	3	style		
line_cap	3	style		
line_join	3	style		
line_width	3	style		
line_miter	3	style		
letter_spacing	3	style		
source_list		object_list		Source list
type		source_list		Type
id		source_list		ID

10.6.4. Responses Example:

Success return data:

{"status":"ok","id":1,"name":"Msg","attribute":{"created_time":1696228686,"modified_time":1696228686,"permission":"rwx","print_log":false,"printdata_pref":{"print_prefs":[{"ff_margin":60,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"

bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false}]}},"style":{"rotate":0,"mirror":0,"stretch":0,"reverse":0},"object_list":[{"type":"text","name":"text","id":1,"attribute":{"locked":false,"lock_aspect_ratio":true,"relative_resize":true,"enabled":true},"style":{"font_style":"ttf-default-r*nnn*-378-378-UTF-8","rotate":0,"mirror":0,"stretch":0,"reverse":0,"visiblity":"visible","x":0,"y":23,"w":777,"h":110,"halign":0,"valign":0,"row_space":1,"letter_space":10,"fh_ratio":0,"fw_ratio":0,"pivot_x":0,"pivot_y":0,"scale_x":1,"scale_y":1,"paint_style":"fill","line_cap":"butt","line_join":"miter","line_width":0,"line_miter":1,"letter_spacing":1},"source_list":[{"type":"text","id":1}]}]}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.7. Get message list

10.7.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/data/list	Interface path, Fixed value
offset	1			Start bit
num	1			totality

10.7.2. Example

{"request_type": "get","path": "/data/list","offset":0,"num":10}

10.7.3. Responses Data Schema

Name	Level	Previous level	Data	Description
data_list	1			data_list
id	2	data_list		ID
name	2	data_list		Name
attribute	2	data_list		
created_time	3	attribute		Creation time
modified_time	3	attribute		Modified time
permission	3	attribute		Permission
printdata_pref	3	attribute		Pint delay
print_prefs	4	printdata_pref		
ff_margin	5	print_prefs		Front delay
fr_margin	5	print_prefs		Rear delay
bf_margin	5	print_prefs		Roundtrip reverse-Front delay
br_margin	5	print_prefs		Roundtrip reverse –Rear delay

continuous_print	5	print_prefs		Continuous print

10.7.4. Responses Example:

Success return data:

{"data_list":[{"id":3,"name":"Msg2","attribute":{"created_time":1696233050,"modified_time":1696233050,"permission":"rwx","printdata_pref":{"print_prefs":[{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false}]}}},{"id":2,"name":"Msg1","attribute":{"created_time":1696233015,"modified_time":1696233015,"permission":"rwx","printdata_pref":{"print_prefs":[{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false}]}}},{"id":1,"name":"Msg","attribute":{"created_time":1696228686,"modified_time":1696228686,"permission":"rwx","printdata_pref":{"print_prefs":[{"ff_margin":60,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false},{"ff_margin":0,"fr_margin":0,"bf_margin":0,"br_margin":0,"continuous_print":false}]}}}]}

10.8. Delete source

10.8.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/data/source	Interface path, Fixed value
id	1			Source id
type	1			Source type

10.8.2. Example

{"request_type": "delete","path": "/data/source","id":1,"type":"text"}

10.8.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

10.8.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.9. Delete object

10.9.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/data/object	Interface path, Fixed value
id	1			Object id

10.9.2. Example

{"request_type": "delete","path": "/data/object","id":1}

10.9.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

10.9.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.10. Delete message

10.10.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		delete	Request type, Fixed value
path	1		/data/data	Interface path, Fixed value
id	1			Message id

10.10.2. Example

{"request_type": "delete","path": "/data/data","id":1}

10.10.3. Responses Data Schema

Name	Level	Previous level	Data	Description

status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

10.10.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

Error return data:

{"status":"Error","code":12295,"descript":"no exist Error"}

10.11. Modify Source

10.11.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		put	Request type, Fixed value
path	1		/data/source	Interface path, Fixed value
id	1			Source id
type	1			Type
name	1			Name
attribute	1			
content	2	attribute		Content

10.11.2. Example

{"request_type": "put","path": "/data/source","id": 1,"type": "text","name": "text","attribute": {"content": "TEXT"}}

10.11.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

10.11.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

10.12. Modift object

10.12.1. Data Schema:

Name	Level	Previous level	Data	Description

request_type	1		put	Request type, Fixed value
path	1		/data/object	Interface path, Fixed value
id	1			Object id
type	1			Type
name	1			Name
style	1			Style
x	2	style		X position
y	2	style		Y position
w	2	style		Width
h	2	style		Height
pivot_x	2	style		X pivot
pivot_y	2	style		Y pivot
rotate	2	style		Rotation
scale_x	2	style		X Scale
scale_y	2	style		Y Scale
paint_style	2	style		Paint style
line_cap	2	style		
line_join	2	style		
line_width	2	style		
miter_limit	2	style		
font_style	2	style		Font style
visiblity	2	style		Visibility
attribute	1			
source_list	1		Array	Source list
type	2	source_list		Source type
id	2	source_list		Source id

10.12.2. Example

{"request_type": "put","path": "/data/object","id": 2,"type": "text","name": "text","style": {"x": 0,"y": 23,"w": 777,"h": 110,"pivot_x": 0,"pivot_y": 0,"rotate": 0.0,"scale_x": 1.0,"scale_y": 1.0,"paint_style": "fill","line_cap": "butt","line_join": "miter","line_width": 0,"miter_limit": 4,"font_style": "ttf-default-r*nnn*-378-378-UTF-8","visiblity": "visible"},"attribute": {},"source_list": [{"type": "text","id": 2}]}

10.12.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

10.12.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

10.13. Modify message

10.13.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/data/data	Interface path, Fixed value
id	1			Message id
type	1			Type
name	1			Name
attribute	1			
printdata_pref	2	attribute		Delay
print_prefs_pref	3	printdata_pref		
ff_margin	4	print_prefs_pref		Front delay
fr_margin	4	print_prefs_pref		Rear delay
bf_margin	4	print_prefs_pref		Roundtrip reverse-Front delay
br_margin	4	print_prefs_pref		Roundtrip reverse –Rear delay
object_list	1			Object list
id	2	object_list		Object id
type	2	object_list		Object type

10.13.2. Example

{"request_type": "put","path": "/data/data","id": 2,"name": "Message","attribute": {"printdata_pref": {"print_prefs": [{"ff_margin": 60.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0 }, {"ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0 },{"ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0},{"ff_margin": 0.0,"fr_margin": 0.0,"bf_margin": 0.0,"br_margin": 0.0}]}},"object_list": [{"id": 2,"type": "text"}]}

10.13.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			

descript	1		ok	conclusion

10.13.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

11. Actively report the print status

11.1. Actively report the print status

11.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/engine/print_status	Interface path, Fixed value
status	1			Status
code	1			Code
descript	1			Descript
index	1			

11.1.2. Example

{"request_type":"post","path":"/engine/print_status","status":"ok","code":0,"descript":"is ok","index":[1,2,2,1]}

11.1.3. Responses Data Schema

12. Send dynamic data

12.1. Send dynamic data

12.1.1. Data Schema:

Name	Level	Previous level	Data	Description
request_type	1		get	Request type, Fixed value
path	1		/info/heart_beat	Interface path, Fixed value

print_mode	1		single:Send one and print one multiple: Send one and keep printing	Print mode
data	1			Data
type	1			Type
name	1			Name
content	1			Content

12.1.2. Example

{"request_type":"post","path":"/engine/dynamic","print_mode":"single","data":[{"type":"text","name":"text","content":"w"},{"type":"text","name":"text1","content":"c"},{"type":"text","name":"text2","content":"w"}]}

12.1.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		ok or Error	Status
code	1			
descript	1		ok	conclusion

12.1.4. Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}

12.2. Send dynamic image

12.2.1. Data Schema：

Name	Level	Previous level	Data	Description
request_type	1		post	Request type, Fixed value
path	1		/engine/download_image	Interface path, Fixed value
parm	1			Binary algorithm parameters
name	1			Source name
size	1			Image size
content	1			Image content(base64 encoding)

12.2.2. Example

{"request_type":"post","path":"/engine/download_image","parm":1,"size":502,"name":"i1","content":"Qk32AQAAAAAAAD4AAAAoAAAANQAAADcAAAABAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAP///wD//wAAf

4AAAP/+AAAfgAAA//gAAA+AAAD/8AAAB4AAAP/gAAADAAAA/+AAAAAAAAD
/wAD8AAAAAP/AA/+AABgA/4AP/+AAOAD/gB//8AA4AP+AH//4ADgA/4A///w
AOAD/AD//+AA4AP8AP//4ADgA/wA///gAOAD/AD//+AA4AP+AP//8ADgA/4A
///wAOAD/gD///AA4AP+AH//8ADgA/8Af//4AOAD/wA///gA4AP/gA///ABgA/
+AA//8AAAD/8AAH/4AAAP/4AAAHgAAA//wAAAAAAAD//gAAAAAAAP//gAAA
AAAA///wAAAAAAD///8AAAA4AP///4AADgA/////wAOAD/////4A4AP/////
/gDgA/////+AOAD/////4A4AP/////gDgA/////8AOAD/////wB4AP/9////A
HgA//z///4AeAD//D///AB4AP/8D//4AHgA//wB//AACAD//AAfgAAAAP/8AAAA
AAAA//wAAAAAAAD//AAAAAAAAP//AAAACAAA///AAAA+AAD///gAAP/AAP/
//4AH//gA///////+AD///////4AA==="}

12.2.3. Responses Data Schema

Name	Level	Previous level	Data	Description
status	1		Ok or error	status
code	1			
descript	1		ok	conclusion

12.2.4 Responses Example:

Success return data:

{"status":"ok","code":0,"descript":"ok"}